GA2 - Temă seminar 1

1. $(A+B)^n = C_n^0 A^n B^0 + C_n^1 A^{n-1} B^1 + \dots + C_n^{n-1} A^1 B^{n-1} + C_n^n A^0 B^n$

cum $A^0 = B^0 = I_n \Rightarrow$

$(A+B)^n = C_n^0 A^n + C_n^1 A^{n-1} B^1 + \dots + C_n^{n-1} A^1 B^{n-1} + C_n^n B^n$

cum $C_n^i = C_n^{n-i}$, $i \in \mathbb{N}$, $i = \overline{0,n} \Rightarrow$

$(A+B)^n = C_n^n A^n + C_n^{n-1} A^{n-1} B^1 + \dots + C_n^1 A^1 B^{n-1} + C_n^0 B^n$

$$= \sum_{j=0}^{n} C_n^j A^j B^{n-j}$$

2. $$\left.\begin{array}{l} A + B = B + A \\ A + B = AB \\ B + A = BA \end{array}\right\| \Rightarrow AB = BA$$

3. $A = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix} \Rightarrow A^n = \begin{pmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{pmatrix}$ $(\forall)\, n \in \mathbb{N}^*$

Aplicăm inducția matematică:

$p(n): A^n = \begin{pmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{pmatrix}$ $(\forall)\, n \in \mathbb{N}^*$

Etapa de verificare

$p(1): A^1 = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix}$ (din ipoteză „A")

$p(2): A^2 = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix}$

$$A^2 = \begin{pmatrix} \cos^2 \alpha - \sin^2 \alpha & 2 \sin \alpha \cos \alpha \\ -2 \sin \alpha \cos \alpha & \cos^2 \alpha - \sin^2 \alpha \end{pmatrix}$$

cum $\begin{cases} \sin 2\alpha = 2 \sin \alpha \cos \alpha \\ \cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha \end{cases} \Rightarrow$

$\Rightarrow A^2 = \begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{pmatrix}$ „A"

Etapa de inducție propriu-zisă:

presupunem $p(k)$, $k \in \mathbb{N}^*$ propoziție adevărată, dem. că $p(k+1)$ prop „A":

$p(k+1): A^{k+1} = A^k \cdot A$

$$A^{k+1} = \begin{pmatrix} \cos k\alpha & \sin k\alpha \\ -\sin k\alpha & \cos k\alpha \end{pmatrix} \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}$$

$$= \begin{pmatrix} \cos k\alpha \cos \alpha - \sin k\alpha \sin \alpha & \cos k\alpha \sin \alpha + \sin k\alpha \cos \alpha \\ -(\cos k\alpha \sin \alpha + \sin k\alpha \cos \alpha & \cos k\alpha \cos \alpha - \sin k\alpha \sin \alpha \end{pmatrix}$$

cum $\begin{cases} \cos(a+b) = \cos a \cos b - \sin a \sin b \\ \sin(a+b) = \sin a \cos b + \cos a \sin b \end{cases} \Rightarrow$

$\Rightarrow A^{k+1} = \begin{pmatrix} \cos(k+1)\alpha & \sin(k+1)\alpha \\ -\sin(k+1)\alpha & \cos(k+1)\alpha \end{pmatrix}$ „A" $\Rightarrow$ inducție completă $\Rightarrow$

$\Rightarrow A^n = \begin{pmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{pmatrix}$, $n \in \mathbb{N}^*$

4. $A = \begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 1 & 3 & 4 \\ 2 & 5 & 1 & -1 \\ -1 & -2 & 2 & 4 \end{pmatrix}$

$$\det A = \begin{vmatrix} 1 & 1 & 2 & 3 \\ 1 & 1 & 3 & 4 \\ 2 & 5 & 1 & -1 \\ -1 & -2 & 2 & 4 \end{vmatrix} \underset{\substack{C_2 = C_2 - C_1 \\ C_3 = C_3 - 2C_1 \\ C_4 = C_4 - 3C_1}}{=\!=\!=} \begin{vmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 2 & 3 & -3 & -7 \\ -1 & -1 & 4 & 7 \end{vmatrix} =$$

$$\underset{\text{dezv } L_1}{=} 1 \cdot (-1)^{1+1} \begin{vmatrix} 0 & 1 & 1 \\ 3 & -3 & -7 \\ -1 & 4 & 7 \end{vmatrix} = 0 + 7 + 12 - 3 - 21 - 0 = -5$$

GA 2 - Temă seminar 2

1. $A = \begin{pmatrix} 1 & 3 & 0 & -2 \\ 2 & 1 & -1 & 3 \\ 1 & 2 & 3 & -1 \\ -1 & 4 & 0 & 1 \end{pmatrix}$

$$\det A = \begin{vmatrix} 1 & 3 & 0 & -2 \\ 2 & 1 & -1 & 3 \\ 1 & 2 & 3 & -1 \\ -1 & 4 & 0 & 1 \end{vmatrix} \underset{\substack{L_2 = L_2 + \frac{3}{2} L_1 \\ L_3 = L_3 - \frac{1}{2} L_1 \\ L_4 = L_{\cancel{3}1} + \frac{1}{2} L_1}}{=\!=} \begin{vmatrix} 1 & 3 & 0 & -2 \\ \frac{7}{2} & \frac{11}{2} & -1 & 0 \\ \frac{1}{2} & \frac{1}{2} & 3 & 0 \\ -\frac{1}{2} & \frac{11}{2} & 0 & 0 \end{vmatrix} =$$

$$\underset{\text{dezv } c_4}{=\!=} (-1)^{1+4} \cdot (-2) \begin{vmatrix} \frac{7}{2} & \frac{11}{2} & -1 \\ \frac{1}{2} & \frac{1}{2} & 3 \\ -\frac{1}{2} & \frac{11}{2} & 0 \end{vmatrix} = 2\left(0 - \frac{33}{4} - \frac{11}{4} - \frac{1}{4} - \frac{231}{4} - 0\right)$$

$$= 2 \cdot \left(-\frac{276}{4}\right) = -\frac{276}{2} = -138$$

2. $A_3(1,2) = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$

$$\left(\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right) \underset{L_2 \Leftrightarrow L_1}{\Rightarrow} \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 \\ 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right)$$

$$\underset{\substack{L_2 = L_2 - 2L_1 \\ L_3 = L_3 - L_1}}{\Rightarrow} \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & -3 & -1 & 1 & -2 & 0 \\ 0 & -1 & 1 & 0 & -1 & 1 \end{array}\right) \underset{\substack{L_3 \Leftrightarrow L_2 \\ L_2 = -L_2}}{\Rightarrow} \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & -3 & -1 & 1 & -2 & 0 \end{array}\right)$$

$$\Rightarrow \quad \begin{vmatrix} \cancel{1} & \cancel{0} & \cancel{3} \\ \cancel{0} & \cancel{-3} & \cancel{3} \\ \cancel{0} & \cancel{0} & \cancel{4} \end{vmatrix} \quad \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & -1 & 2 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & 0 & -4 & 1 & 1 & -3 \end{array}\right) \Rightarrow$$

$L_1 = L_1 - 2L_2$

$L_3 = L_3 + 3L_1$

$$\underset{L_4 = -\frac{L_4}{4}}{\Rightarrow} \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & -1 & 2 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & +\frac{3}{4} \end{array}\right) \underset{\substack{L_2 = L_2 + L_3 \\ L_1 = L_1 - 3L_3}}{\Rightarrow} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ 0 & 1 & 0 & -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{array}\right)$$

$$\Rightarrow A_3(1,2)^{-1} = \begin{pmatrix} \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{pmatrix}$$

3. $$\Delta_1 = \begin{vmatrix} x & a & a & a \\ a & x & a & a \\ a & a & x & a \\ a & a & a & x \end{vmatrix} = 0 \quad \Longrightarrow$$

$L_1 = L_1 + L_2 + L_3 + L_4$

$$\Delta_1 = \begin{vmatrix} x+3a & x+3a & x+3a & x+3a \\ a & x & a & a \\ a & a & x & a \\ a & a & a & x \end{vmatrix} =$$

$$= (x+3a) \begin{vmatrix} 1 & 1 & 1 & 1 \\ a & x & a & a \\ a & a & x & a \\ a & a & a & x \end{vmatrix} \underset{\substack{C_2 = C_2 - C_1 \\ C_3 = C_3 - C_1 \\ C_4 = C_4 - C_1}}{=\!=} (x+3a) \begin{vmatrix} 1 & 0 & 0 & 0 \\ a & x-a & 0 & 0 \\ a & 0 & x-a & 0 \\ a & 0 & 0 & x-a \end{vmatrix}$$

$$\underset{\text{dezv } L_1}{=\!=} (x+3a) \cdot 1 \cdot (-1)^{1+1} \begin{vmatrix} x-a & 0 & 0 \\ 0 & x-a & 0 \\ 0 & 0 & x-a \end{vmatrix} = (x+3a)(x-a)^3$$

$$(x+3a)(x-a)^3=0 \Rightarrow \begin{cases} x_1=-3a \\ x_2=x_3=x_4=a \end{cases}$$

4. $$\Delta_n = \begin{vmatrix} -1 & a & a & \dots & a \\ a & -1 & a & \dots & a \\ a & a & -1 & \dots & a \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a & a & a & \dots & -1 \end{vmatrix} \underset{\substack{L_2=L_2-L_1 \\ L_3=L_3-L_1 \\ \vdots \\ L_n=L_n-L_1}}{=} \begin{vmatrix} -1 & a & a & \dots & a \\ a+1 & -(a+1) & 0 & \dots & 0 \\ a+1 & 0 & -(a+1) & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a+1 & 0 & 0 & \dots & -(a+1) \end{vmatrix} =$$

$$\underset{C_1=C_1+C_2+\dots+C_n}{=} \begin{vmatrix} (n-1)a-1 & a & a & \dots & a \\ 0 & -(a+1) & 0 & \dots & 0 \\ 0 & 0 & -(a+1) & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & -(a+1) \end{vmatrix} \underset{\text{dezv } C_1}{=}$$

$$((n-1)a-1)\cdot(-1)^{1+1} \begin{vmatrix} -(a+1) & 0 & \dots & 0 \\ 0 & -(a+1) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -(a+1) \end{vmatrix} =$$

$$= ((n-1)a-1)\cdot(-1)^{n-1}\cdot(a+1)^{n-1}$$

5. $$\Delta_2 = \begin{vmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_1 \\ x_3 & x_1 & x_2 \end{vmatrix}$$

x_1, x_2, x_3 răd ec $x^3-2x^2+2x+17=0$

$s_1 = x_1+x_2+x_3 = 2$

$s_2 = x_1x_2+x_1x_3+x_2x_3 = 2$

$S_2 = x_1^2+x_2^2+x_3^2 = s_1^2-2s_2 = 4-4 = 0$

$$\Delta_2 \underset{c_1 = c_1 + c_2 + c_3}{=} (x_1+x_2+x_3)\begin{vmatrix} 1 & x_2 & x_3 \\ 1 & x_3 & x_1 \\ 1 & x_1 & x_2 \end{vmatrix} = (x_1+x_2+x_3)(x_2x_3 + x_1x_2 + x_1x_3 - x_3^2 - x_2^2 - x_1^2)$$

$$\Delta_2 = \underbrace{(x_1+x_2+x_3)}_{s_1}(\underbrace{x_1x_2 + x_1x_3 + x_2x_3}_{s_2} - \underbrace{(x_1^2+x_2^2+x_3^2)}_{S_1})$$

$$\Delta_2 = 2 \cdot (2 - 0) = 4$$

6\. a) $$\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ yz & xz & xy \end{vmatrix} \overset{?}{=} (x-y)(y-z)(z-x)(xy+xz+yz)$$

$$\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ yz & xz & xy \end{vmatrix} \underset{\substack{c_2 = c_2 - c_1 \\ c_3 = c_3 - c_1}}{=} \begin{vmatrix} x & y-x & z-x \\ x^2 & y^2-x^2 & z^2-x^2 \\ yz & z(x-y) & y(x-z) \end{vmatrix} =$$

$$= (y-x)(z-x)\begin{vmatrix} x & 1 & 1 \\ x^2 & y+x & z+x \\ yz & -z & -y \end{vmatrix} \underset{c_1 = c_1 - xc_2}{=} (y-x)(z-x)\begin{vmatrix} 0 & 1 & 1 \\ -xy & y+x & z+x \\ z(y+x) & -z & -y \end{vmatrix}$$

$$\underset{c_2 = c_2 - c_3}{=} (y-x)(z-x)\begin{vmatrix} 0 & 0 & 1 \\ -xy & y-z & z+x \\ z(y+x) & -(z-y) & -y \end{vmatrix} = (y-x)(z-x)(y-z)\begin{vmatrix} 0 & 0 & 1 \\ -xy & 1 & z+x \\ z(y+x) & -1 & -y \end{vmatrix}$$

$$\underset{\text{dezv } L_1}{=} (y-x)(z-x)(y-z) \cdot 1 \cdot (-1)^{1+3}\begin{vmatrix} -xy & 1 \\ z(y+x) & +1 \end{vmatrix} =$$

$$= -(x-y)(y-z)(z-x)(-xy - zy - zx) = (x-y)(y-z)(z-x)(xy+zy+zx)$$

b) $\begin{vmatrix} a+b & b+c & c+a \\ a^2+b^2 & b^2+c^2 & c^2+a^2 \\ a^3+b^3 & b^3+c^3 & c^3+a^3 \end{vmatrix} = 2abc(a-b)(b-c)(c-a)$

$$\begin{vmatrix} a+b & b+c & c+a \\ a^2+b^2 & b^2+c^2 & c^2+a^2 \\ a^3+b^3 & b^3+c^3 & c^3+a^3 \end{vmatrix} \underset{\substack{C_2=C_2-C_1 \\ C_3=C_3-C_1}}{=} \begin{vmatrix} a+b & c-a & c-b \\ a^2+b^2 & c^2-a^2 & c^2-b^2 \\ a^3+b^3 & c^3-a^3 & c^3-b^3 \end{vmatrix} =$$

$$=(c-a)(c-b)\begin{vmatrix} a+b & 1 & 1 \\ a^2+b^2 & c+a & c+b \\ a^3+b^3 & c^2+ca+a^2 & c^2+cb+b^2 \end{vmatrix} \underset{\substack{C_1=C_1-aC_2- \\ -bC_3}}{=} \begin{vmatrix} 0 \\ -c(a+b) \\ -ca(a+c)-cb(b+c) \end{vmatrix}$$

$$= (c-a)(c-b)\begin{vmatrix} 0 & 1 & 1 \\ -c(a+b) & c+a & c+b \\ -ca(a+c)-cb(b+c) & c^2+ca+a^2 & c^2+cb+b^2 \end{vmatrix} \underset{C_2=C_2-C_3}{=}$$

$$= -c(c-a)(c-b)\begin{vmatrix} 0 & 0 & 1 \\ +(a+b) & a-b & c+b \\ a(a+c)+b(b+c) & c(a-b)+a^2-b^2 & c^2+cb+b^2 \end{vmatrix} \underset{dezv\, r_1}{=}$$

$$= -c(c-a)(c-b)\cdot 1\cdot(-1)^{1+1}\begin{vmatrix} +(a+b) & a-b \\ a(a+c)+b(b+c) & c(a-b)+a^2-b^2 \end{vmatrix} =$$

$$= -c(a-b)(c-a)(c-b)\begin{vmatrix} +(a+b) & 1 \\ a(a+c)+b(b+c) & a+b+c \end{vmatrix}$$

$$= -c(a-b)(c-a)(c-b)(a^2+b^2+2ab+\cancel{ac}+\cancel{bc}-a^2-b^2-\cancel{ac}-\cancel{bc})$$

$$= 2abc(a-b)(b-c)(c-a)$$

7. a) $\Delta_3 = \begin{vmatrix} 1 & 3 & 0 \\ 0 & 1 & 3 \\ 3 & 0 & 1 \end{vmatrix} = 1 + 3^3 + 0 - 0 - 0 - 0 = 28$

$$\Delta_4 = \begin{vmatrix} 1 & 3 & 0 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 3 \\ 3 & 0 & 0 & 1 \end{vmatrix} \underset{L_4 = L_4 - 3L_1}{=} \begin{vmatrix} 1 & 3 & 0 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & -9 & 0 & 1 \end{vmatrix} \underset{dezv\ c_1}{=}$$

$$= 1 \cdot (-1)^{1+1} \begin{vmatrix} 1 & 3 & 0 \\ 0 & 1 & 3 \\ -9 & 0 & 1 \end{vmatrix} = 1 - 3^4 + 0 - 0 - 0 - 0 = 1 - 81 = -80$$

b) $$\Delta_n = \begin{vmatrix} 1 & 3 & 0 & \dots & 0 & 0 \\ 0 & 1 & 3 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 3 \\ 3 & 0 & 0 & \dots & 0 & 1 \end{vmatrix} \underset{dezv\ c_1}{=}$$

$$= 1 \cdot (-1)^{1+1} \begin{vmatrix} 1 & 3 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{vmatrix} + 3 \cdot (-1)^{n+1} \begin{vmatrix} 3 & 0 & \dots & 0 \\ 1 & 3 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 3 \end{vmatrix}$$

$$= 1 \cdot (-1)^2 \cdot 1^{n-1} + (-1)^{n+1} \cdot 3^n$$

$$= 1 + (-1)^{n+1}\, 3^n$$

8. a) $\Delta_n = \begin{vmatrix} 4 & 3 & 0 & \dots & 0 \\ 1 & 4 & 3 & \dots & 0 \\ 0 & 1 & 4 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 4 \end{vmatrix} \underset{\text{dezv } L_1}{=}$

$= 4 \cdot (-1)^{1+1} \underbrace{\begin{vmatrix} 4 & 3 & \dots & 0 \\ 1 & 4 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 4 \end{vmatrix}}_{\Delta_{n-1}} + 3 \cdot (-1)^{1+2} \begin{vmatrix} 1 & 3 & 0 & \dots & 0 \\ 0 & 4 & 3 & \dots & 0 \\ 0 & 1 & 4 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & 3 \\ 0 & 0 & 0 & \dots 1 & 4 \end{vmatrix} =$

$= 4\Delta_{n-1} - 3 \cdot 1 \cdot (-1)^{1+1} \underbrace{\begin{vmatrix} 4 & 3 & \dots & 0 \\ 1 & 4 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 4 \end{vmatrix}}_{\Delta_{n-2}} = 4\Delta_{n-1} - 3\Delta_{n-2}$ $(\forall)\, n \geq 3$, $n \in \mathbb{N}$

b) $\Delta_n - 4\Delta_{n-1} + 3\Delta_{n-2} = 0$

$r^2 - 4r + 3 = 0$, $r_1 = 1$, $r_2 = 3$ $\Rightarrow$ $\Delta_n = A r_1^n + B r_2^n$

$\Delta_n = A \cdot 1^n + B \cdot 3^n$

$= 3^n B + A$; A, B const

$\Delta_3 = \begin{vmatrix} 4 & 3 & 0 \\ 1 & 4 & 3 \\ 0 & 1 & 4 \end{vmatrix} = 64 - 12 - 12 = 40$

$\Delta_4 = \begin{vmatrix} 4 & 3 & 0 & 0 \\ 1 & 4 & 3 & 0 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 1 & 4 \end{vmatrix} \underset{L_1 = L_1 - 4L_2}{=} \begin{vmatrix} 0 & -13 & -12 & 0 \\ 1 & 4 & 3 & 0 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 1 & 4 \end{vmatrix} \underset{\text{dezv } c_1}{=}$

$$\Delta_4 = 1\cdot(-1)^{1+2}\begin{vmatrix} -13 & -12 & 0 \\ 1 & 4 & 3 \\ 0 & 1 & 4 \end{vmatrix} = -(208 - 0 + 0 - 0 + 39 + 48) = 121$$

$$\begin{cases} 27B + A = 40 \\ 81B + A = 121 \end{cases} \Rightarrow 54B = 81 \Rightarrow B = \frac{81}{54} = \cancel{\frac{9}{6}}\ \frac{3}{2} \Rightarrow A = -\frac{1}{2}$$

$$\Rightarrow \Delta_n = \frac{3}{2}\cdot 3^n - \frac{1}{2} = \frac{3^{n+1}-1}{2} \quad (\forall)\ n \ge 3,\ n \in \mathbb{N}$$

GA 2 - Tema seminar 3

1. a) $S_1 \begin{cases} x + y - z = 2 \\ 2x + y - 3z = 2 \\ x - y - z = 0 \end{cases}$

$$A^e = \left(\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 2 & 1 & -3 & 2 \\ 1 & -1 & -1 & 0 \end{array}\right) \underset{\substack{L_2 = L_2 - 2L_1 \\ L_3 = L_3 - L_1}}{=} \left(\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -1 & -1 & -2 \\ 0 & -2 & 0 & -2 \end{array}\right) \underset{L_2 = L_2 \cdot (-1)}{=}$$

$$= \left(\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & -2 & 0 & -2 \end{array}\right) \underset{\substack{L_1 = L_1 - L_2 \\ L_3 = L_3 + 2L_2}}{=} \left(\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 2 & 2 \end{array}\right) \underset{\substack{\cancel{L_1 = L_1 + L_3} \\ L_3 = \frac{L_3}{2}}}{=}$$

$$= \left(\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right) \underset{\substack{L_1 = L_1 + 2L_3 \\ L_2 = L_2 - L_3}}{=} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right) \Rightarrow \begin{cases} x = 2 \\ y = 1 \\ z = 1 \end{cases}$$

S_1 sis com det $\Rightarrow S'_1 = \{(2, 1, 1)\}$

b) $S_2 \begin{cases} x + y + 2z = 1 \\ x + y - 3z = 1 \\ x + y - 2z = 1 \end{cases}$

$$A^e = \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 1 & 1 & 3 & 1 \\ 1 & 1 & -2 & 1 \end{array}\right) \underset{\substack{L_2 = L_2 - L_1 \\ L_3 = L_3 - L_1}}{=} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & \boxed{0} & \boxed{1} & 0 \\ 0 & 0 & -4 & 0 \end{array}\right)$$

avem 0 pe al doilea pivot (poz (2, 2)) $\Rightarrow$ sistemul este incompatibil (nu are soluții). $\Rightarrow S'_2 = \emptyset$

c) $S_3 \begin{cases} x + y + z + t = 1 \\ 2x - y + z - t = 2 \\ x - 2y + 0z - 2t = -1 \end{cases}$

$$A^e = \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 2 & -1 & 1 & -1 & 2 \\ 1 & -2 & 0 & -2 & -1 \end{array}\right) \underset{\substack{L_2 = L_2 - 2L_1 \\ L_3 = L_3 - L_1}}{=} \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & -3 & -1 & -3 & 0 \\ 0 & -3 & -1 & -3 & -2 \end{array}\right) =$$

$$\underset{L_2 = \frac{L_2}{-3}}{=} \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & \frac{1}{3} & 1 & 0 \\ 0 & -3 & -1 & -3 & -2 \end{array}\right) \underset{\substack{L_1 = L_1 - L_2 \\ L_3 = L_3 + 3L_2}}{=} \left(\begin{array}{cccc|c} 1 & 0 & \frac{2}{3} & 0 & 1 \\ 0 & 1 & \frac{1}{3} & 1 & 0 \\ 0 & 0 & \boxed{0} & \boxed{0} & -2 \end{array}\right)$$

al treilea pivot o să fie pe coloana termenilor liberi => sistemul S_3 este incompatibil $S_3 = \emptyset$

d) $S_4 \begin{cases} x + 2y + 3z - 2t = 6 \\ 2x - y - 2z - 3t = 8 \\ 3x + 2y - z + 2t = 4 \\ 2x - 3y + 2z + t = -8 \end{cases}$

$$A^e = \left(\begin{array}{cccc|c} 1 & 2 & 3 & -2 & 6 \\ 2 & -1 & -2 & -3 & 8 \\ 3 & 2 & -1 & 2 & 4 \\ 2 & -3 & 2 & 1 & -8 \end{array}\right) \underset{\substack{L_2 = L_2 - 2L_1 \\ L_3 = L_3 - 3L_1 \\ L_4 = L_4 - 2L_1}}{=} \left(\begin{array}{cccc|c} 1 & 2 & 3 & -2 & 6 \\ 0 & -5 & -8 & +1 & -4 \\ 0 & -4 & -10 & 8 & -14 \\ 0 & -7 & -4 & 5 & -20 \end{array}\right)$$

$$\underset{\substack{L_2 = \frac{L_2}{-5} \\ L_3 = \frac{L_3}{2}}}{=} \left(\begin{array}{cccc|c} 1 & 2 & 3 & -2 & 6 \\ 0 & 1 & \frac{8}{5} & -\frac{1}{5} & \frac{4}{5} \\ 0 & -2 & -5 & 4 & -7 \\ 0 & -7 & -4 & 5 & -20 \end{array}\right) \underset{\substack{L_1 = L_1 - 2L_2 \\ L_3 = L_3 + 2L_2 \\ L_4 = L_4 + 7L_2}}{=} \left(\begin{array}{cccc|c} 1 & 0 & -\frac{1}{5} & -\frac{8}{5} & \frac{22}{5} \\ 0 & 1 & \frac{8}{5} & -\frac{1}{5} & \frac{4}{5} \\ 0 & 0 & -\frac{9}{5} & \frac{18}{5} & -\frac{27}{5} \\ 0 & 0 & \frac{36}{5} & \frac{18}{5} & -\frac{72}{5} \end{array}\right) =$$

$$\underset{\substack{L_3 = L_3 \cdot -\frac{5}{9} \\ L_4 = L_4 \cdot \frac{5}{16}}}{=} \left(\begin{array}{cccc|c} 1 & 0 & \frac{1}{5} & -\frac{8}{5} & \frac{22}{5} \\ 0 & 1 & \frac{8}{5} & \frac{1}{5} & \frac{4}{5} \\ 0 & 0 & 1 & -2 & 3 \\ 0 & 0 & 2 & 1 & -4 \end{array}\right) \underset{\substack{L_1 = L_1 + \frac{L_3}{5} \\ L_2 = L_2 - L_3 \cdot \frac{8}{5} \\ L_4 = L_4 - 2L_3}}{=}$$

$$= \left(\begin{array}{cccc|c} 1 & 0 & 0 & -2 & 5 \\ 0 & 1 & 0 & 3 & -4 \\ 0 & 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 5 & 10 \end{array}\right) \underset{L_4 = \frac{L_4}{5}}{=} \left(\begin{array}{cccc|c} 1 & 0 & 0 & -2 & 5 \\ 0 & 1 & 0 & 3 & -4 \\ 0 & 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right) \underset{\substack{L_1 = L_1 + 2L_4 \\ L_2 = L_2 - 3L_4 \\ L_3 = L_3 + 2L_4}}{=}$$

$$= \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right) \Rightarrow \begin{cases} x = 1 \\ y = 2 \\ z = -1 \\ t = -2 \end{cases}$$

S_4 is comp det $\Rightarrow S_4 = \{(1, 2, -1, -2)\}$

2.a) $S_1 \begin{cases} x + y - z = 0 \\ 2x + y - 3z = 0 \\ x - y - z = 0 \end{cases}$

$$A^e = \left(\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 2 & 1 & -3 & 0 \\ 1 & -1 & -1 & 0 \end{array}\right) \underset{\substack{L_2 = L_2 - 2L_1 \\ L_3 = L_3 - L_1}}{=} \left(\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & -2 & 0 & 0 \end{array}\right) \underset{L_2 = -L_2}{=}$$

$$= \left(\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -2 & 0 & 0 \end{array}\right) \underset{\substack{L_1 = L_1 - L_2 \\ L_3 = L_3 + 2L_2}}{=} \left(\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right) \underset{L_3 = \frac{L_3}{2}}{=}$$

$$= \left(\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right) \underset{\substack{L_1 = L_1 + 2L_3 \\ L_2 = L_2 - L_3}}{=} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

S_1 sis comp det $\Rightarrow S'_1 = \{(0,0,0)\}$

b)

$$S_2 \begin{cases} x + y - z + t = 0 \\ x - y + z + t = 0 \\ 2x + y + 2z - t = 0 \end{cases}$$

$$A^e = \left(\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 0 \\ 1 & -1 & 1 & 1 & 0 \\ 2 & 1 & 2 & -1 & 0 \end{array}\right) \underset{\substack{L_2 = L_2 - L_1 \\ L_3 = L_3 - 2L_1}}{=} \left(\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 0 \\ 0 & -2 & 2 & 0 & 0 \\ 0 & -1 & 4 & -3 & 0 \end{array}\right)$$

$$\underset{L_2 = \frac{L_2}{-2}}{=} \left(\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & -1 & 4 & -3 & 0 \end{array}\right) \underset{\substack{L_1 = L_1 - L_2 \\ L_3 = L_3 + L_2}}{=} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 3 & -3 & 0 \end{array}\right)$$

$$\underset{L_3 = \frac{L_3}{3}}{=} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \end{array}\right) \underset{L_2 = L_2 + L_3}{=} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & -1 & 0 \end{array}\right)$$

x, y, z var principale

$t = \alpha$, $\alpha \in \mathbb{R}$ var sec

S_2 sis comp ned $\Rightarrow \begin{cases} x = -\alpha \\ y = \alpha \\ z = \alpha \end{cases}$

$\Rightarrow S'_2 = \{(-\alpha, \alpha, \alpha, \alpha) \mid \alpha \in \mathbb{R}\}$

3. a) $A = \left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{array}\right) \underset{L_1 \Leftrightarrow L_2}{=} \left(\begin{array}{cc|cc} 1 & 3 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array}\right) \underset{L_2 = L_2 - 2L_1}{=}$

$= \left(\begin{array}{cc|cc} 1 & 3 & 0 & 1 \\ 0 & -5 & 1 & -2 \end{array}\right) \underset{L_2 = \frac{L_2}{-5}}{=} \left(\begin{array}{cc|cc} 1 & 3 & 0 & 1 \\ 0 & 1 & -\frac{1}{5} & +\frac{2}{5} \end{array}\right) \underset{L_1 = L_1 - 3L_2}{=}$

$= \left(\begin{array}{cc|cc} 1 & 0 & \frac{3}{5} & -\frac{1}{5} \\ 0 & 1 & -\frac{1}{5} & \frac{2}{5} \end{array}\right) \Rightarrow A^{-1} = \begin{pmatrix} \frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{pmatrix}$

b) $B = \left(\begin{array}{ccc|ccc} 2 & -1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \underset{L_1 \Leftrightarrow L_2}{=} \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 0 & 1 & 0 \\ 2 & -1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \underset{\substack{L_2 = L_2 - 2L_1 \\ L_3 = L_3 - L_1}}{=}$

$= \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 0 & 1 & 0 \\ 0 & -3 & -3 & 1 & -2 & 0 \\ 0 & 0 & -1 & 0 & -1 & 1 \end{array}\right) \underset{L_2 = \frac{L_2}{-3}}{=} \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 0 & 1 & 0 \\ 0 & +1 & +1 & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & -1 & 0 & -1 & 1 \end{array}\right) \underset{L_1 = L_1 - L_2}{=}$

$= \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 1 & 1 & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & -1 & 0 & -1 & 1 \end{array}\right) \underset{L_3 = -L_3}{=} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 1 & 1 & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & -0 & 1 & 0 & 1 & -1 \end{array}\right) \underset{\substack{L_1 = L_1 - L_3 \\ L_2 = L_2 - L_3}}{=}$

$= \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{3} & -\frac{2}{3} & 1 \\ 0 & 1 & 0 & -\frac{1}{3} & -\frac{1}{3} & 1 \\ 0 & 0 & 1 & 0 & 1 & -1 \end{array}\right) \Rightarrow B^{-1} = \begin{pmatrix} \frac{1}{3} & -\frac{2}{3} & 1 \\ -\frac{1}{3} & -\frac{1}{3} & 1 \\ 0 & 1 & -1 \end{pmatrix}$

1) $C = \left(\begin{array}{ccc|ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 2 & 1 & -2 & 0 & 1 & 0 \\ 1 & -1 & 4 & 0 & 0 & 1 \end{array}\right) \underset{\substack{L_2 = L_2 - 2L_1 \\ L_3 = L_3 - L_1}}{=} \left(\begin{array}{ccc|ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 0 & -1 & -8 & -2 & 1 & 0 \\ 0 & -2 & 1 & -1 & 0 & 1 \end{array}\right)$

$\underset{L_2 = -L_2}{=} \left(\begin{array}{ccc|ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 0 & 1 & 8 & 2 & -1 & 0 \\ 0 & -2 & 1 & -1 & 0 & 1 \end{array}\right) \underset{\substack{L_1 = L_1 - L_2 \\ L_3 = L_3 + 2L_2}}{=} \left(\begin{array}{ccc|ccc} 1 & 0 & -5 & -1 & 1 & 0 \\ 0 & 1 & 8 & 2 & -1 & 0 \\ 0 & 0 & 17 & 3 & -2 & 1 \end{array}\right)$

$\underset{L_3 = \frac{L_3}{17}}{=} \left(\begin{array}{ccc|ccc} 1 & 0 & -5 & -1 & 1 & 0 \\ 0 & 1 & 8 & 2 & -1 & 0 \\ 0 & 0 & 1 & \frac{3}{17} & -\frac{2}{17} & \frac{1}{17} \end{array}\right) \underset{\substack{L_1 = L_1 + 5L_3 \\ L_2 = L_2 - 8L_3}}{=}$

$= \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{2}{17} & \frac{7}{17} & \frac{5}{17} \\ 0 & 1 & 0 & \frac{10}{17} & -\frac{1}{17} & -\frac{8}{17} \\ 0 & 0 & 1 & \frac{3}{17} & -\frac{2}{17} & \frac{1}{17} \end{array}\right) \Rightarrow C^{-1} = \begin{pmatrix} -\frac{2}{17} & \frac{7}{17} & \frac{5}{17} \\ \frac{10}{17} & -\frac{1}{17} & -\frac{8}{17} \\ \frac{3}{17} & -\frac{2}{17} & \frac{1}{17} \end{pmatrix}$

CAL. - Temă seminar 4

1\. a) $S_3 = \{ v_1 = (1,1,0),\ v_2 = (1,0,1),\ v_3 = (0,1,1) \} \subset \mathbb{R}^3/\mathbb{R}$

$S_3 \subset \mathbb{R}^3/\mathbb{R}$ sis de gen $\Leftrightarrow (\forall)\ v \in \mathbb{R}^3,\ (\exists)\ \alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$

a.î. $v = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$

Fie $v = (x, y, z) \in \mathbb{R}^3$
vector arbitrar

$v = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 \Leftrightarrow$

$(x, y, z) = (\alpha_1 + \alpha_2,\ \alpha_1 + \alpha_3,\ \alpha_2 + \alpha_3)$

$\begin{cases} \alpha_1 + \alpha_2 = x \\ \alpha_1 + \alpha_3 = y \\ \alpha_2 + \alpha_3 = z \end{cases} \Rightarrow 2(\alpha_1 + \alpha_2 + \alpha_3) = x + y + z \Rightarrow \begin{cases} \cancel{\alpha_1 = x + y} \\ \cancel{\alpha_2 = x + z} \\ \cancel{\alpha_3 = y + z} \end{cases}$

$\Rightarrow \alpha_1 + \alpha_2 + \alpha_3 = \dfrac{x + y + z}{2} \Rightarrow \begin{cases} \alpha_1 = \dfrac{x + y - z}{2} \\ \alpha_2 = \dfrac{x - y + z}{2} \\ \alpha_3 = \dfrac{-x + y + z}{2} \end{cases}$

deci $(\exists)$ $\begin{cases} \alpha_1 = \dfrac{x + y - z}{2} \\ \alpha_2 = \dfrac{x - y + z}{2} \\ \alpha_3 = \dfrac{-x + y + z}{2} \end{cases}$ a.î. $(\forall)\ v = (x, y, z) \in \mathbb{R}^3 \Rightarrow$

$v = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$

$\Rightarrow S_3 = \{ v_1, v_2, v_3 \} \subset \mathbb{R}^2/\mathbb{R}$ sist de gen pt $\mathbb{R}^2/\mathbb{R}$

b) $S_4' = \{v_1 = 1,\ v_2 = X-1,\ v_3 = (X-1)^2\} \subset \mathbb{R}_2[X]$

$S_4 \subset \mathbb{R}_2[X]$ sis de gen $\Leftrightarrow$ $(\forall)\ p \in \mathbb{R}_2[X]$, $(\exists)\ \alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$

$$p = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$$

Fie $p = aX^2 + bX + c \in \mathbb{R}_2[X]$
polinom arbitrar

$p = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 \Leftrightarrow$

$aX^2 + bX + c = \alpha_3 X^2 + (\alpha_2 - 2\alpha_3)X + \alpha_1 - \alpha_2 + \alpha_3$

$$\begin{cases} a = \alpha_3 \\ b = \alpha_2 - 2\alpha_3 \\ c = \alpha_1 - \alpha_2 + \alpha_3 \end{cases} \Rightarrow \quad \begin{aligned} &\alpha_3 = a \\ &\alpha_2 = b + 2a \\ &\alpha_1 = c + b + 2a - a = a + b + c \end{aligned}$$

deci $(\exists)$ $\begin{cases} \alpha_3 = a \\ \alpha_2 = 2a + b \\ \alpha_1 = a + b + c \end{cases}$ a.î. $(\forall)\ p = aX^2 + bX + c \in \mathbb{R}_2[X]$
$p = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 \Rightarrow$

$\Rightarrow S_4' = \{v_1, v_2, v_3\} \subset \mathbb{R}_2[X]$ sis de gen (pt $\mathbb{R}_2[X]$)

GAL. - Temă seminar 5

1. $U = \{(x, y, z) \in \mathbb{R}^3 \,/\, -x + 3y + z = 0\} \subset \mathbb{R}^3/\mathbb{R}$

a) Fie $v_1, v_2 \in U$ $\Rightarrow$ $v_1 = (x_1, y_1, z_1)$, $-x_1 + 3y_1 + z_1 = 0$

$\alpha_1, \alpha_2 \in \mathbb{R}$ $\quad v_2 = (x_2, y_2, z_2)$, $-x_2 + 3y_2 + z_2 = 0$

$\alpha_1 v_1 + \alpha_2 v_2 \in U$?

~~$\alpha_1 v_1 + \alpha_2 v_2 = (-\alpha_1 x_1 - \alpha_2 x_2, 3\alpha_1 y_1 + 3\alpha_2 y_2, \alpha_1 z_1 + \alpha_2 z_2)$~~

$\alpha_1 v_1 + \alpha_2 v_2 = (\underbrace{\alpha_1 x_1 + \alpha_2 x_2}_{x}, \underbrace{\alpha_1 y_1 + \alpha_2 y_2}_{y}, \underbrace{\alpha_1 z_1 + \alpha_2 z_2}_{z})$

$-x + 3y + z = -\alpha_1 x_1 - \alpha_2 x_2 + 3\alpha_1 y_1 + 3\alpha_2 y_2 + \alpha_1 z_1 + \alpha_2 z_2$

$= \alpha_1 (\underbrace{-x_1 + 3y_1 + z_1}_{0}) + \alpha_2 (\underbrace{-x_2 + 3y_2 + z_2}_{0})$

$= 0 \Rightarrow \alpha_1 v_1 + \alpha_2 v_2 \in U \Rightarrow U \subset \mathbb{R}^3$ sp vect

b) Fie $v \in U$ vector arbitrar, $v = (x, y, z)$, $-x + 3y + z = 0$

$x = 3y + z$

$U \ni v = (3y + z, y, z) = (3y, y, 0) + (z, 0, z)$

$= y\underbrace{(3, 1, 0)}_{u_1} + z\underbrace{(1, 0, 1)}_{u_2} = y u_1 + z u_2 \Rightarrow$

$S' = \{u_1, u_2\} \subset U$ sis de gen (1)

fie $\alpha_1, \alpha_2 \in \mathbb{R}$ a.î. $\alpha_1 u_1 + \alpha_2 u_2 = 0_{\mathbb{R}^3}$

$\alpha_1 (3, 1, 0) + \alpha_2 (1, 0, 1) = (0, 0, 0)$

$$\Rightarrow \begin{cases} 3\alpha_1 + \alpha_2 = 0 \\ \alpha_2 = 0 \\ \alpha_1 = 0 \end{cases} \Rightarrow \alpha_1 = 0,\ \alpha_2 = 0 \text{ sol unică} \Rightarrow$$

$\Rightarrow S' = \{u_1, u_2\} \subset \mathbb{R}^3$ s.v. lin indep (2)

$\overset{(1),(2)}{\Rightarrow} \underset{\text{bază}}{S' \subset U} \Rightarrow \dim_{\mathbb{R}} U = 2$

G.A.2. - Temă seminar 6

1. $V_1 = \{(x, y, 0) / x, y \in \mathbb{R}\}$

$V_2 = \{(u, 0, v) / u, v \in \mathbb{R}\}$

a) Fie $w_1, w_2 \in V_2 \Rightarrow w_1 = (u_1, 0, v_1), \quad u_1, v_1 \in \mathbb{R}$

$\alpha_1, \alpha_2 \in \mathbb{R} \qquad w_2 = (u_2, 0, v_2) \quad u_2, v_2 \in \mathbb{R}$

$\alpha_1 w_1 + \alpha_2 w_2 = (\alpha_1 u_1 + \alpha_2 u_2, 0, \alpha_1 v_1 + \alpha_2 v_2)$

$\alpha_1 u_1 + \alpha_2 u_2 \in \mathbb{R}$

$\alpha_1 v_1 + \alpha_2 v_2 \in \mathbb{R}$

$\Rightarrow \alpha_1 w_1 + \alpha_2 w_2 \in V_2 \Rightarrow V_2 \subseteq \mathbb{R}^3$ sp vect

$V_2 \ni (u, 0, v) = u e_1 + v e_3 \Rightarrow B_2 = \{e_1, e_3\}$ bază

$u, v \in \mathbb{R}$

$\Rightarrow \dim V_2 = 2$ (pl ~~vertical)~~ vectorial)

b) T dimensiunii (Grossmann)

$\dim (V_1 + V_2) = \dim V_1 + \dim V_2 - \dim V_1 \cap V_2$

$V_1 \cap V_2 \ni v \Rightarrow \begin{cases} x = u \ (\overset{not}{=} t) \\ y = 0 \\ v = 0 \end{cases} \Rightarrow V_1 \cap V_2 = \{(t, 0, 0) / t \in \mathbb{R}\}$

$= \{t e_1 / t \in \mathbb{R}\} = \langle e_1 \rangle$

$\Rightarrow \dim V_1 \cap V_2 = 1$

$\dim V_1 = \dim V_2 = 2$

$\Rightarrow \dim (V_1 + V_2) = 2 + 2 - 1 = 3$

$V_1 + V_2 \subseteq \mathbb{R}^3$ sp vect

$\Rightarrow V_1 + V_2 = \mathbb{R}^3$

$V_1 \oplus V_2 = \mathbb{R}^3$ nu este adevărată, deoarece $V_1 \cap V_2 = \langle e_1 \rangle \neq 0_{\mathbb{R}^3}$

2. $V_1 = \{A \in M_n(\mathbb{R}) \ / \ Tr\, A = 0\}$

$V_2 = \{A \in M_n(\mathbb{R}) \ / \ A = x I_n, \ x \in \mathbb{R}\}$

a) fie $A_1, A_2 \in V_1$ $\Rightarrow$ $Tr\, A_1 = 0$

$\alpha_1, \alpha_2 \in \mathbb{R}$ $\quad Tr\, A_2 = 0$

$tr(\alpha_1 A_1 + \alpha_2 A_2) = Tr(\alpha_1 A_1) + Tr(\alpha_2 A_2) =$

$= \alpha_1 Tr\, A_1 + \alpha_2 Tr\, A_2 = 0 + 0 = 0 \Rightarrow \alpha_1 A_1 + \alpha_2 A_2 \in V_1$

deci $V_1 \subset M_n(\mathbb{R})$
sp vect

fie $A_3, A_4 \in V_2$ $\Rightarrow$ $A_3 = \lambda_1 I_n, \ \lambda_1 \in \mathbb{R}$

$\alpha_3, \alpha_4 \in \mathbb{R}$ $\quad A_4 = \lambda_2 I_n, \ \lambda_2 \in \mathbb{R}$

$\alpha_3 A_3 + \alpha_4 A_4 = \alpha_3 \lambda_1 I_n + \alpha_4 \lambda_2 I_n =$

$= (\alpha_3 \lambda_1 + \alpha_4 \lambda_2) I_n, \ \alpha_3 \lambda_1 + \alpha_4 \lambda_2 \in \mathbb{R} \Rightarrow \alpha_3 A_3 + \alpha_4 A_4 \in V_2$

deci $V_2 \subset M_n(\mathbb{R})$
sp vect

b) $V_1 + V_2 \subset M_n(\mathbb{R})$, dem că $M_n(\mathbb{R}) \subset V_1 + V_2$

$(\forall)\, C \in M_n(\mathbb{R})$, $(\exists)\, A \in V_1$, $B \in V_2$ a.î. $C = A + B$

notăm $\alpha = \frac{tr\, C}{n} \in \mathbb{R}$

$A = C - \alpha I_n$

evident: $Tr\, A = Tr\, C - Tr(\alpha I_n) = Tr\, C - \frac{Tr\, C}{n} \cdot n = 0$

$B = \alpha I_n$

luăm: $A = C - \alpha I_n$

$B = \alpha I_n$

deci $(\forall) C \in \mathcal{M}_n(\mathbb{R})$, $(\exists)$ $A \in V_1$, $B \in V_2$ a.î. $C = A + B \Rightarrow$

$\mathcal{M}_n(\mathbb{R}) \subset V_1 + V_2 \Rightarrow \mathcal{M}_n(\mathbb{R}) = V_1 + V_2$

arătăm că $V_1 \cap V_2 = \{O_n\}$

fie $D \in V_1 \cap V_2 \Rightarrow \left.\begin{array}{l} D = x I_n \\ \operatorname{tr} D = 0 \end{array}\right| \Rightarrow x = 0 \Rightarrow D = O_n$

deci $V_1 \oplus V_2 = \mathcal{M}_n(K)$

c) verif Th dimensiunii:

$\dim_{\mathbb{R}} \mathcal{M}_n(\mathbb{R}) = \dim_{\mathbb{R}} V_1 + \dim_{\mathbb{R}} V_2$

$\dim_{\mathbb{R}} \mathcal{M}_n(\mathbb{R}) = n^2$

fie $A \in V_1$, $A = (a_{ij})_{i,j = \overline{1,n}}$, $\operatorname{Tr} A = 0$

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \\ a_{n1} & a_{n2} & & -a_{11} - \dots - a_{n-1\,n-1} \end{pmatrix} = \sum_{1 \le i,j \le n} a_{ij} E'_{ij}, \text{ unde:}$$

$$E'_{ij} = \begin{cases} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & & -1 \end{pmatrix}, & \text{pt } i = j \neq n \text{: } 1 \text{ pe poz } (i,j),\ -1 \text{ pe poz } (n,n),\ 0 \text{ pe celelalte poz} \\[2ex] \begin{pmatrix} 0 & \dots 1 \dots & 0 \\ 0 & 0 & \\ 0 & \dots & 0 \end{pmatrix}, & \text{pt } i \neq j \text{: } 1 \text{ pe poz } (i,j),\ 0 \text{ pe celelalte poz} \\[2ex] \cancel{\begin{pmatrix} 0 & \dots & 0 \\ \vdots & & \\ 0 & & 0 \end{pmatrix} = O_n, \text{ pt } i = j = n,\ 0 \text{ pe toate poz}} \end{cases}$$

$B_{V_1} = \{E'_{ij}\}_{1 \le i,j \le n} \subset V_1$
bază

$\dim V_1 = \text{card } B_{V_1} = n^2 - 1$

fie $B \in V_2$, $B = (b_{ij})_{1 \le i,j \le n}$, $B = \lambda I_n$, $\lambda \in \mathbb{R}$

$$B = \begin{pmatrix} \lambda & 0 & \dots & 0 \\ 0 & \lambda & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda \end{pmatrix} = \lambda E''_{ij}, \text{ unde } E''_{ij} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

$B_{V_2} = \{E''_{ij}\} \subset V_2$
bază

$\dim V_2 = \text{card } B_{V_2} = 1$

avem: $n^2 - 1 + 1 = n^2$ (verif Th dimen)

3. $f: \mathbb{R}^3 \to \mathbb{R}^3$
$f(x, y, z) = (x + y + z, x - y + z, x - y - z)$

a) $f(X) = AX$, unde $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ și $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{pmatrix} \in \mathcal{M}_3(\mathbb{R})$

fie $X_1, X_2, X_3 \in \mathbb{R}^3$
$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$ $\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$ $\begin{pmatrix} x_3 \\ y_3 \\ z_3 \end{pmatrix}$

și $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$

$f(\alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_3) = A(\alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_3) =$
$= A(\alpha_1 X_1) + A(\alpha_2 X_2) + A(\alpha_3 X_3) = (A\alpha_1) X_1 + (A\alpha_2) X_2 + (A\alpha_3) X_3 =$

$= (\alpha_1 A) x_1 + (\alpha_2 A) x_2 + (\alpha_3 A) x_3 = \alpha_1 (A x_1) + \alpha_2 (A x_2) + \alpha_3 (A x_3) =$

$= \alpha_1 f(x_1) + \alpha_2 f(x_2) + \alpha_3 f(x_3) \rightarrow f$ apl lin

b) $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{pmatrix}$ matr asoc apl lin f în rap cu bazele canonice din $\mathbb{R}^3$, resp $\mathbb{R}^3$

$\begin{array}{ccc} \shortparallel & \shortparallel & \shortparallel \\ f(e_1) & f(e_2) & f(e_3) \end{array}$, unde $\{e_1, e_2, e_3\} \subset \mathbb{R}^3$ b canonică

$f(e_1) = f(1, 0, 0) = (1, 1, 1)$

$f(e_2) = f(0, 1, 0) = (1, -1, -1)$

$f(e_3) = f(0, 0, 1) = (1, 1, -1)$